

CEO Accountability for Corporate Lawsuits

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Abstract

Corporate lawsuits have been found to depress stock performance and undermine corporate reputation. I draw on agency theory and two competing theoretical arguments to test whether CEOs should be held accountable for corporate lawsuits. By examining lawsuit cases of Chinese listed firms from 1999 to 2011, I first document that information asymmetry makes firms more likely to get involved in lawsuits, confirming the assertion from theoretical literature. I find that the incidence of lawsuits increases the propensity of CEO turnover in defendant firms regardless of lawsuit results. In plaintiff firms, such sensitivity between lawsuit and CEO turnover becomes lower if they win. Finally, I show that board independence and private controlling shareholders strengthen CEO accountability of plaintiff firms. The results are robust to the control of relevant firm characteristics, performance and governance measures and are consistent after controlling for selection bias by applying propensity score matching. This paper offers insights of effective functioning of internal governance mechanisms in China following corporate litigations.

Keywords: litigation, defendant, plaintiff, court verdict, CEO, turnover, board, independence, financial distress, information asymmetry, SOE, China

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1. Introduction

Many studies examine corporate lawsuits in the U.S. and document a variety of discomfoting impacts on firms' well-being including negative share performance, tarnished corporate reputation and constrained business opportunities (Culter and Summers, 1988; Englemann and Cornell, 1988; Bhagat, Brickley, and Coles, 1994, 1998; Muoghalu, Robison, and Glascock, 1990; Hersch, 1991; Koku, Qureshi, and Akhigbe, 2001; Cumming, Haslem and Knill, 2011). Firth et al. (2011) empirically set their research in China, where legal system is perceived as under-developed (Allen, Qian and Qian, 2005) and highly "capricious" (Firth et al., p. 573, 2011), and they document similar results that litigation negatively affects the share performance for both the defendant and plaintiff upon announcement, with financial distressed defendants suffering from even worse economic consequences. Despite the richness of literature on this issue, however, there have been few studies considering the consequences of corporate lawsuits to corporate governance of the litigant firms, such as the turnover of the top management team. This study thus fills this gap in the literature and sheds light on this important but severely under-researched topic.

This paper takes the agency perspective argued by Fama and Jensen (1983). Due to the separation of ownership and control, the top managers are the ones who make decisions of firm's daily operations. Therefore one can reasonably expect corporate lawsuits would be a result of these decisions either directly or through indirect ways. If the internal monitoring mechanism is effective and if there is anyone who should be held accountable for the series of negative consequences following corporate litigations, the management would be on the top of the list.

Anecdotal evidence on management turnover following corporate lawsuits is mixed. In some cases, the CEOs are fired after the revelation of corporate litigation. For example,

Xinjiang Hops' then CEO, Aikelamu Aishayoufu, was fired following a series of corporate litigations over breach of loan contracts in 2005. Qin Gang, the former CEO of Sichuan Mingxing Electric Power, unexpectedly resigned in four months after the announcement of a corporate lawsuit regarding the quality of product. The CEO of Sichuan Western Resources Holding, Wang Yong, was also ousted in the following months after a lawsuit case involving the breach of a loan covenant. In other cases, the lawsuit cases did not trigger the change of the management team. Such examples include China Minsheng Banking, China Southern Airlines and Weifang Beida Jadebird Huaguang Technology, in which CEOs stayed at the company even after a serious of corporate litigations.

On the one hand, there are some reasons that corporate lawsuits may prompt management turnover. First, upon the announcement of litigations, firms' share prices are found to be depressed (Firth et al., 2011). As argued by Weisbach (1988) and Warner et al. (1988), poor share performance increases the probability of management turnover. Second, corporate litigations in general are very costly (legal consulting fees and expenses to hire lawyers etc.) and in some cases the costs of corporate lawsuits are extremely large (Culter and Summers, 1988). Therefore firms that are exposed to lawsuits are more likely to suffer from financial distress. As stated by Gilson (1989), firms with financial distress have higher propensity to experience management turnover than their counterparts. Lastly, corporate lawsuits have significantly negative impacts on firm reputation (Haslem, 2005; Englemann and Cornell, 1988). To regain its reputation, firms may be more inclined to change their management (Agrawal et al., 1999)

On the other hand, there are also various reasons that corporate lawsuits may not lead to management turnover. To begin with, corporate internal control mechanisms in general are insufficient to trigger value-adding management changes (Jensen, 1993; Agrawal et al., 1999). This phenomenon is even more likely to be the case in China where corporate governance is

weak (Allen et al., 2005; Firth et al., 2011). Second, the cost to replace a fired CEO's firm-specific human capital may outweigh its benefits. As a result, sacking top managers may not be an optimal choice for litigant firms (Agrawal and Cooper, 2007). Third, the costs to establish an internal control mechanism that could completely eliminate litigation risk are extremely high. So current exposure to corporate litigations may not trigger the change of the management team. Finally, while corporate lawsuits are generally bad news for firms, in some cases firms may not suffer from reputational loss upon the litigation. For example, firms may initiate a lawsuit against others without concrete evidence of any wrongdoings. In this case, the benefits of change its management may be negligible.

As illustrated above, the top management team of firms could experience greatly different consequences upon the corporate lawsuits. In this paper, I examine empirically the effects of corporate lawsuits on subsequent changes of firms' top management, namely CEOs. In doing so, I try to test the above competing theoretical arguments about the consequences to the top managers when firms are involved in a corporate lawsuit.

To conduct the research, I construct a sample of corporate lawsuit cases filed in China from 1999 to 2011 in which at least one side, plaintiff or defendant, is listed in Chinese stock markets. In total, I identify 3,301 cases² over this period. Firstly, I test the determinants of corporate litigation, with particular interest in how state ownership and information asymmetry affect the probability of corporate lawsuits. In China, the state still has dominant authority over many aspects (Hou and Moore, 2011; Firth, 2011). Particularly germane to this context is its ability to exert pervasive pressure on courts' decision (Allen et al., 2005; Chen, 2003). I therefore use a dummy variable of state-owned enterprise (SOE) to examine the state-ownership effects. Although literature abounds with a variety of theories and models that stress

² These include different types of litigation, such as breach of contract, breach of loan covenants, breach of loan guarantee agreement, intangible disputes, financial service disputes etc.

the importance of information asymmetry on the probability of litigation (Bebchuk, 1984; Reinganum and Wilde, 1986; Png, 1987; Schweizer, 1989), few study takes a further step to empirically test it. Using analyst coverage as proxy of information asymmetry (Clarke and Shastri, 2000; Chae, 2005; Uysal, Kedia and Panchapagesan, 2008), I assess what role information asymmetry between litigants has been playing in determining the likelihood of a lawsuit. The results lend evidence to the information asymmetry theory that higher analyst coverage, a proxy of lower information asymmetry, is correlated with lower probability of firm initiating a lawsuit after controlling for various firm characteristics. My evidence also suggests that SOEs are indeed favoured in the judicial process after controlling for the same set of factors. Particularly, for defendant, I document that a company is less likely to be sued if it is state-owned. For plaintiff, the evidence is even stronger that being an SOE increases the probability of winning a legal case. These result are in line with Firth et al. (2011), confirming that state-owned firms fare better in the judicial process.

To test the central research questions, I identify 1,512 forced CEO turnovers in Chinese listed companies from 1999 to 2011. However, to verify whether a CEO is held accountable for corporate litigation, I need to address a potential endogeneity problem, which may stem from the sample selection process. Specifically, factors, such as performance and board characteristics, that could cause a firm being involved in corporate lawsuits may also cause CEO replacement. To overcome this issue, I follow the literature (Rosenbaum and Rubin, 1983, 1985) and apply propensity-score matching (PSM) method to construct a control group (i.e., firms not being involved in litigation), wherein each company is an otherwise closest match with a company from the treatment group (i.e., firms that are involved in litigation) except for the treatment factor (being involved in lawsuits). I then apply a probit model with forced CEO turnover as the dependant variable and corporate lawsuits as independent variable to examine the main question in both the original and the matching samples. The results provide *prima facie* evidence that an increase in the probability of forced CEO turnover can

be observed in both plaintiff and defendant firms, with more pronounced effects witnessed in defendant firm. These results exist in both the original sample and the PSM control sample and are robust to the controls of various firm characteristics (e.g., firm size, stock and operating performance, board size, supervisory board size, leverage, financial distress, etc.) as well as industry and year effects. I also document that SOEs are slack at disciplining their chief executives when their shareholders' value is harmed, which undermines the internal monitoring mechanism in state-owned enterprises. Board characteristics, such as duality, do matter when it comes to disciplining CEOs. Financial distress, as proxied by "Special Treatment"³ (ST), is also supported by the evidence to be an influential determinant in CEO turnover. After adding the interaction effects between litigation and SOE, results further confirm my propositions.

This study makes important contributions to the literature in several ways. This is the first study to examine the effectiveness of the internal monitoring mechanism as to whether CEOs are held accountable for corporate litigations across various types of legal disputes, with the largest sample so far. Drawing on agency theory and the two competing theoretical arguments mentioned above, I propose that CEOs should (or should not) be held accountable for such events where shareholders' value is severely harmed due to the negative consequences following litigation. As significant improvements in listed firms' monitoring system have been documented in recent years (Firth et al. 2005; Hou, 2014), this paper presents insights on how the impact of these changes enhance the quality of monitoring mechanism in China by examining arguably the most important monitoring results, CEO turnover. In addition, this is also the first study, to the best of my knowledge, to empirically test the effects of information asymmetry on the probability of corporate lawsuits. I document *prima facie* evidence that

³ In a typical stock delisting process, the stock exchanges will first label a firm in financial trouble as a special treatment firm (ST), then designate it a particular transfer (PT) firm if it fails to turn profitable within one year (Liu and Lu, 2007).

lower information asymmetry is associated with lower probability of firm filing a lawsuit. Then, I add new evidence on the debate whether litigation can serve as an effective instrument of corporate governance. Finally, I also lend support to that SOEs are indeed favoured in the judicial process.

The reminder of this article is organized as follows. Section 2 presents the institutional background. Section 3 reviews the literature and develops the testable hypotheses. Section 4 introduces the research design followed by a discussion of empirical results in section 5. Section 6 then concludes.

2. Institutional Background

2.1 Legal System in China

The legal system in China is categorized as a large civil law system (Firth, et al., 2011). Politically, as the devolution of power exaggerated, the old justice system that was based on political ideology to exert control was incapable of forging regularity to government operations. Specifically, some policies within the old system are even contradict to each other. In order to bring stability and impose unity on the process of governance, a new legal system was introduced to replace the old one. In the economic realm, it was also initially intended to formalize only state-owned entities (SOEs), creating an efficiency-driven mechanism among them (Clarke et al., 2006).

The Third Plenum of the Eleventh Central Committee of the Communist Party in 1978, which officially jettisoned class struggle as the central task of the Party and embraced the carriage of “socialist market economy” steered by Deng Xiaoping, marks the watershed of the legal system in China. From that point, where law was endowed with another important role,

encouraging private economic activity and fostering entrepreneurship, legal reform started. In the span of two decades into the reform, the National People's Congress (NPC) and its Standing Committee have passed about 160 laws (Cai, 1999). Among them, on July 1st, 1979, the Law on Sino-Foreign Equity Joint Venture aimed at attracting foreign investment into China was passed; on December 4th, 1982, the Constitution was completely reconstructed; and on April 12th, 1986, General Principles of Civil Law were passed. Even the long-delayed property rights and human rights have also been officially recognized in the constitution since 2004 (Clarke et al., 2006). The state council and local administrative did not remain stagnant. Over 5900 local and administrations decrees have also been issued over the same period (Cai, 1999). Coupled with domestic economic development and further opening up to foreign investments, the Communist Party led by President Xi Jinping is now strengthening the rule of law in China. At the annual plenum meeting that just ended in October, 2014, the party made it clear that "Socialist rule of law with Chinese characteristics" would be fully implemented by 2020 and the change would be "extensive and profound"⁴.

2.2 Government Intervention

Although the legal system has been improved enormously in the past decades, it is still plagued by the lack of independence and impartiality owing to the heavy political influence exerted by the ruling party and local governments (Zhang, 2003). The fundamental reason for the phenomenon can be explained by a brief introduction of the Chinese administrative hierarchy and relationship network between the government and the courts, as the courts are a key factor in forming an effective legal system where law can be turned into social reality. In China, at any administrative level, the People's Congress has the ultimate authority to appoint and dismiss the leader of the court, and the local governments also have control over court's finances and logistics (Zhang, 2003; Clark et al., 2008). In other words, the welfare of the

⁴ See "China with legal characteristics", 1st November 2014, the Economist.

officials of the court and their families practically rests on the hands of the People's Congress and local governments. Further, the general principle applies in China that the administrative rank of the leadership has dominant influence on the rank of the institution he or she leads. Although the head of court at any given level in China is not necessarily subordinate to the head of government at that given level, the administrative rank of the court's leader is typically lower than the official rank of the head of the local government. As a result, it is very difficult, if not impossible, for courts to remain independent and autonomous even should they wish to do so (Zhang, 2003; Clark et al. 2008).

As an example of the weak and low status of courts in China's political system, when an enforcement of judgement involving cooperation of banks is issued by a people's court at any given level, this instruction must be signed not only by the court of issuance but also by the regulator of the bank at that given level if it intends to be effectively implemented (Zhang, 2003). Although the courts are given the authority and jurisdiction to issue the judgement, its implementation cannot always be guaranteed.

Despite the large expansion of the legal system since the outset of the reform, its substance for the government and the political party is largely unaltered. It is still an important tool of governance and control and essentially remains government-driven, not citizen- or litigant-driven.

2.3 Shortage of Legal Professionals

A sufficient supply of qualified legal professionals is another precondition to form an effective legal system. In 1979, the organizational structure of courts and procurators' offices were reconstructed to better serve the subjects of economic reform. By 1997, the total number of people employed in various court and procurators' offices had grown to over 500,000.

Meanwhile, the number of lawyers in China had expanded to 100,000 (Zhang, 2003). By 2005, almost three decades into the reform, the number of lawyers had increased to about 119,000⁵, approximate 14 times higher than the number in 1983 (Chen, 2004). However, although the number of lawyers has increased substantially over time, the number of lawyers in China is only about the same as the number of qualified attorneys in a single state in the U.S. In addition, lawyers are used in only about 50% of criminal lawsuits and even worse, only 10-25% of civil cases are represented by lawyers. Further, only 20% of all lawyers in China have law degrees and most of the lawyers are war veterans (Allen et al., 2005). Clearly, China needs a plethora of time until its supply of qualified legal professionals is ready.

2.4 Conflicts between Different Rules

Another drawback of Chinese legal system is the lack of a clear guidance or institutional features that could be used in identifying the ambiguities and providing reference when conflict between different rules arises. These institutional features are important because of the wide spectrum of bodies that are empowered to make rules with different degrees of binding effects. To illustrate, NPC and its standing committee are at the top of the pyramid of formal lawmaking structure where they are empowered to pass statutes. The second tier is the local governments which have the power to enact administrative regulations in support of the constitution and legislative objectives. The next level is the People's Congress at provincial level and large cities. They have the authority to enact local regulations that are aimed at governing local issues. Standing at the lowest level of this pyramid is the ministries of the local government. The rules issued by them cannot be enforced by courts (Zhang, 2003). When conflict between these rules arises, there is no formal system or any guidance that could resolve the conflict. Although courts prefer to follow the higher-level rules, they have to also uphold the conflicting lower-level rules due to the lack of power to invalidate any legislation. This

⁵ This includes both 103,000 full time and 16,000 part time law practitioners (Ministry of Justice, 2005).

creates inconsistencies in the legal system (Zhang, 2003). For this reason, similar legal cases often have different court verdicts and judge's personal preferences are inevitable involved in the judicial process.

2.5 Corruption

Corruption of public officials has become commonplace in China as both its variety and quantity increased dramatically after the inauguration of the market reform in 1978 (Zhang, 2003). The judicial system is not immune to this phenomenon. These corruptions include, but not limited to, judges entering illegal relationship with litigants for mutual benefits, exchange of favours between judge and litigants, courts colluding with local governments and judges taking bribery, which is the most common one. In March 2012, for example, four judges at Fujian province Taijiang district's People's Court were sentenced to imprisonment of 3 to 11 years for taking bribes of hundreds of thousands renmibi (RMB) (People's Daily, 2014). From March to July 2014, in a local People's Court located in Liaoning province, where the judicial system is relatively impartial compared to an average inland province, ten judges were under investigation for various unlawful activities (People's Daily 2014).

Observers usually agree on two aspects about judiciary corruption. On one hand, corruption has seriously undermined the integrity of law, thus damaging the confidence people have in the legal system. On the other hand, corruption could lead to more litigations being initiated (Firth et al., 2011). Statistics show that between 1993 and 1997, only a span of four years, there were over 370 legal cases where judges were found guilty of crimes (Tan, 1998). In the year of 2000 alone, this number had increased to over 1,290 (People's Daily 2001).

2.6 Number of Lawsuits

The number of lawsuits has increased dramatically during the past few decades. According to the National Statistics Bureau of China, shown in Appendix B, the total number of cases accepted by the courts expanded approximate 14 folds from 513,789 in 1979 to 7,596,116 in 2011. The number of civil litigations increased even faster, being about 16 times larger in 2011 than in 1979. This manifests the increased importance of courts, and to a broader extent, the legal system in people's lives today. As President Xi Jinping is now strengthening the "Socialist rule of law" in China, it also illustrates that courts are regarded by a wider spectrum of audiences as their final resolution to resolve disputes, which warrants research germane to this field, such as this one.

2.7 Relationship between Legal System and Corporate Governance

Corporate governance is closely related with the law and its enforcement. In traditional finance, securities are recognized as a series of cash flows (Modigliani and Miller, 1958). As the understanding of securities improves, Hart (1995) shows a better picture. He argues that the defining features of each kind of securities are not only the cash flows but also the rights attached to them (Hart, 1995). Building on this view, La Porta et al. (1996) in their ground-breaking paper states that these attached rights closely depend on the legal environment of the jurisdictions where these securities are issued and so do the corporate governance. For example, if the legal protection of the investor rights is weak, then the rights of shareholders to vote out unqualified directors and those of the debt holders to claim their investments upon firm's bankruptcy are never guaranteed. As a result, the monitoring role played by investors is severely discouraged. If the law protecting investors' rights is strong, then stakeholders would be more active in terms of monitoring the firm. As argued by Romano (1991), shareholder litigation can serve as an effective governance mechanism in the United States where, as La Porta et al (1996) suggest, the legal protections on shareholder rights are relatively strong.

3. Literature and Hypotheses

3.1 Effects of Information Asymmetry on Corporate Lawsuits

Literature abounds with models that explain what determines the probability of filing a lawsuit and settling the case (Shavell, 1981a and 1981b; Landes, 1971). However only a few of them explicitly take the issue of information asymmetry between the involved parties into account. For example, Bebchuk (1984) examines the probability of litigation and settlement under the backdrop of imperfect information. Based on the assumptions that both the plaintiff and defendant are risk-neutral and the defendant is the party with private information, Bebchuk (1984) develops a model that identifies various factors that influence the probability of settlement including the monetary amount at stake, litigants' legal costs and the nature of both parties' information. He then states that high level of information asymmetry is responsible for the possible failure of parties to settle (Bebchuk, 1984). In other words, high level of information asymmetry increases the probability of filing a lawsuit.

Under the backdrop of "insurance crisis" where potential injurers are subject to a negligence standard and they differ in terms of the cost of taking care of others, Png (1987) utilizes a model to examine the incentives of lawsuit, settlement, and trial as well as the incentives for care and participation in the events that may ultimately cause harm to others. This model is built on several key assumptions including both plaintiff and defendant are risk neutral and the court determines defendant's degree of care without error. Png (1987) finds evidence that all potential defendants, no matter what degree of care they take, experience litigation risk due to information asymmetries in the event of an accident. Because the plaintiff or the victim is apprehensive about whether the defendant has actually taken due care, this information asymmetry is likely to lead them to go file a lawsuit against the potential defendant. Once again, information asymmetry here increases the probability of a litigation.

Many other studies also take information asymmetry into consideration in the study of lawsuit and settlement (Reinganum and Wilde, 1986; Scheweizer, 1989). To empirically test the effects of information asymmetry on litigation, I thus propose the hypothesis as follows:

H1: Firms with low (high) information asymmetry are less (more) likely to be involved in litigation either as defendant or plaintiff.

3.2 Effects of State Ownership on Corporate Lawsuits

The economic reform that started from 1978 also witnessed a fundamental change in the organization structure. The conversion of state-owned enterprises into publicly listed firms was the most prominent one during the reform period. Firth et al. (2006) state that a major characteristic in the enterprise reform was the state retaining controlling ownership in those listed firms. On average, approximately 30% of stake is owned directly by the central and local governments as well as their associated ministries. Another 30% of shares are owned by legal organizations and most of them are ultimately controlled by the state (Chen et al., 2006). Under the backdrop that the Chinese courts are partially dependent on the state and government, Firth et al. (2011) examine whether there is a bias towards politically powerful firms when they are involved in the judicial process. They document that while all firms suffer from negative abnormal returns upon the announcement of corporate litigation, the politically-powerful firms do enjoy some advantages in terms of better stock performance (Frith et al., 2011). Clarke et al. (2006) provide a chronical illustration of Chinese legal system and assert that the political essence of the legal system remains the same as at the outset of the economic reform---a governance tool of the Communist party. Potter (1999) and MacNeil (2002) also assert that it is a tradition for courts to favour the party in which political interest is involved. Drawing on these studies, I aim to empirically examine whether a firm's political power affects its behaviour in the judicial system and I therefore propose the hypothesis as follows:

H2: Politically powerful firms are more (less) likely to sue (be sued by) other companies.

3.3 Outcomes of Corporate Litigation

Studies that focus particularly on stock market reactions to corporate litigations include Bhagat, Brickley, and Coles (1994), who are among the very first to analyse a large inter-firm sample consisting of various types of litigation in the U.S. Their sample covers the period of 1981-1983 and includes 355 firms and 550 observations. Evidence in their study reveals that for defendant firms the realized negative abnormal share return of roughly 1% is significant at 1% level upon filing, while for plaintiff firms the share return is of little significance around lawsuit filings. These results are consistent with findings by Karpoff and Lott (1999). Further, when the combined wealth effects of both defendant and plaintiff firms are taken into consideration, a substantial and statistically significant joint loss is witnessed upon the initiation of lawsuits, which is consistent with the evidence documented in Culter and Summers (1988)'s study.

In a companion study of Bhagat, Brickley and Coles (1994), Bhagat et al. (1998) investigate a larger sample that includes both the inter-firm lawsuits used in the previous study as well as the ones in which opponents are government entities and individuals. They find evidence that heterogeneity measured by different types of legal case and opponent does help explain cross-sectional differences in stock performance around filing and settlement of the lawsuit case. Specifically, defendants suffer from larger negative share returns when government entities stand on the other side of the court than when other types of plaintiffs initiate the lawsuits. Further, evidence also reveals that certain types of litigation, such as environmental, product liability and violation of securities laws are more costly for defendants as compared to legal issues involving anti-trust, breach of contract, and other disputes. Moreover, consistent with

the evidence presented by Lott et al. (1999), Bhagat et al. (1998) find the share performance of defendant upon filing to be positively correlated with company size, after controlling for the types of opposing party and legal cases.

Research that concentrates on a specific type of lawsuits includes Cutler and Summers (1988), who conduct a case study of the Texaco- Pennzoil litigation. Cutler and Summers (1988)'s research provides the first analysis of changes to shareholder wealth in response to corporate litigation, but the study is based on only one litigation case filed for breach of contract. Nevertheless their findings are generally consistent with the evidence presented by Bhagat et al. (1994, 1998) and Koku et al. (2001).

Muoghalu, Robison, and Glascock (1990) use Dodd and Warner's (1983) version of event study methodology to investigate the effects that environmental litigation has on shareholder wealth. The sample consists of 128 lawsuits filed during 1977-1986. They find that shares react negatively to the announcement of filing with a statistically significant mean loss of 1.2% in market value. Contrary to what Bhagat et al. (1994) find, Muoghalu et al. (1990) find no significant abnormal share returns at the disposition of the suit. Lott, Karpoff, and Rankine (1999) uncover similar results that initial settlement announcement is associated with negligible and statistically insignificant average abnormal share returns. Furthermore, Lott et al. (1999) also reveal that stockholders of defendant firms suffer from larger losses should the alleged violations lead to product recalls or assignments of liability for environmental harm. Moreover, they conclude that share value losses are also higher when the plaintiff is the state or the local government.

Hersch (1991) applies a standard "event-study" methodology pioneered by Fama et al. (1969) to a sample of 260 legal cases filed over the period 1964-1986. The sample is homogeneous in a sense that all the cases are related with violating equal employment

opportunity (EEO). He finds that shareholders of these firms generally suffer from a loss in response to the announcement of litigation and the negative abnormal share returns range from -0.294% right after the announcement to a total of 15.6% loss over the 61-day holding period window distributed around the date of initiating class-action lawsuits.

Firth, Rui and Wu (2011) provide the first large-sample study of the wealth effects of corporate litigation in China. By hand collecting, a sample of 1,242 lawsuits filed over the period of 1999-2005 is constructed. They then use standard event study methodology to investigate how litigation news influences the stock performances of litigants. Consistent with evidence from the U.S. (Cutler and Summers, 1988; Hersch, 1991; Bhagat, Brickley, and Coles, 1994; Koku, Qureshi, and Akhigbe, 2001), Firth et al. (2011) find that when litigation is initiated, share price for defendant firm decreases significantly by approximately 1.67%, and this negative effect is more pronounced when the defendant is in poor financial condition. However, contradicting to evidence found in the U.S. market, they reveal that plaintiff firms also experience significantly negative abnormal returns (about 0.77%) on their common stocks.

Haslem (2005) uses a sample of 737 cases to assess how firms with different levels of agency conflicts behave around the initiation and disposition of corporate litigation. He states that the costs of corporate lawsuits not only include the explicit ones, such as legal fees and potential penalties, but also the implicit ones, such as loss of managerial focus as well as corporate reputation and goodwill. Englemann and Cornell (1988) give one possible explanation for the reputation damage that litigation may cause to defendant. Specifically, they state that the suit may convey the notion to the defendant's trading partners that the firm is willing to engage in illegal practices and this characteristic may lead to discomfoting result when it comes to honouring trading contracts. Further, for the lost defendant, the court may impose trading constrains that further damage the firm's value.

Because of the negative abnormal share return, tarnished corporate reputation and constrained business opportunities, firm's value as well as shareholders' interest for both defendant and plaintiff firms is likely to be severely damaged when a firm gets involved in a lawsuit. Taking the standard agency perspective of separation of ownership and control (Fama and Jensen, 1983), the top managers are the ones who make decisions of firm's daily operations. Therefore one can reasonably expect corporate lawsuits as well as those negative consequences would be a result of these decisions either directly or through indirect ways. If the internal monitoring mechanism is effective and if there is anyone who should be held accountable for the series of negative consequences following corporate litigations, the management would be on the top of the list.

Drawing on agency theory (Fama and Jensen, 1983), I propose the following two competing theoretical arguments. On the one hand, there are some reasons that these negative consequences may prompt management turnover. First, as argued by Weisbach (1988) and Warner et al. (1988), poor share performance increases the probability of management turnover. Second, Gilson (1989) states firms with financial distress have higher propensity to experience management turnover than their counterparts. Lastly, to regain firms' reputation that was lost upon the announcement of the corporate lawsuits, firms may be more inclined to change their management (Agrawal et al., 1999)

However, there are also various reasons that corporate lawsuits may not lead to management turnover. To begin with, corporate internal control mechanisms in general are insufficient to trigger value-adding management changes (Jensen, 1993; Agrawal et al., 1999). This phenomenon is even more likely to be the case in China where corporate governance are found to be weak (Allen et al., 2005; Firth et al., 2011). Second, the cost to replace a fired CEO's firm-specific human capital may be outweigh its benefits (Agrawal and Cooper, 2007).

As a result, sacking top managers may be not an optimal choice for litigant firms. Third, the cost to establish an internal control mechanism that could eliminate future litigation risk is extremely high. So current exposure to corporate litigations may not trigger the change of the management team. Finally, while corporate lawsuits are generally bad news for firms, in some cases firms may not suffer from enough reputation loss to trigger a management change. For example, firms may initiate a lawsuit against others without concrete evidence of any wrongdoings. In this case, the benefits of change its management may be negligible. Drawing on agency theory and the above two competing arguments, I thus propose the main hypothesis:

H3a: The probability of forced CEO turnover is higher for litigant firms than for their non-litigant matches.

H3b: The probability of forced CEO turnover is no difference for litigant firms than for their non-litigant matches.

Cutler and Summers (1988) document that 2/3 of the total lost value from both defendant and plaintiff firms was recovered when the Texaco-Pennzoil litigation was finally resolved. Bhagat et al. (1994) also document a positive and significant abnormal share return for defendant firms should litigants settle the case. This suggests that how the litigation is resolved may have an impact on the share value, thus influencing the board's decision to fire the CEO. As a result, I propose the next hypothesis:

H4: For both the litigant firms, winning (not winning) the lawsuit decreases (increases) the probability of forced CEO turnover.

As argued by Jensen and Ruback (1983), poor managers who continue to stay and run the company may be the costliest manifestation of the agency problem. In an ideal environment, underperforming CEOs would be replaced immediately by the board of directors after

pursuing activities where shareholders' interests are damaged. However, anecdotal and empirical evidence suggests that sometimes the reality is different from what normally expected. In firms with a well-functioning and effective board, the unfit managers are replaced very soon to better align shareholder's interests, while in firms with a less effective board, these managers can still operate the business for a rather prolonged period of time. I therefore propose the following hypothesis:

H5: Sensitivities of CEO turnover to corporate litigation are positively associated with board effectiveness.

Firth et al. (2011) state that firms that are plagued by financial distress or close to it may be damaged by a larger magnitude in corporate lawsuits. Several other studies also contend that the most important factor in a litigation is the cost of financial distress (Cutler and Summers, 1988; Bhagat, Brickley and Coles, 1994). Therefore, I also test whether CEOs from financial distressed firms following lawsuits are more likely to be sacked since the damage on firms are more severe. Thus I propose the following hypothesis:

H6: The presence of financial distress enhances the turnover-lawsuits sensitivities.

4. Research Design

4.1 Data and Sample

The past decade witnessed the development of finance and law databases in China. One of the widely used databases, China Centre for Economic Research (CCER/Sinofin), includes lawsuits data such as company status (e.g., defendant, plaintiff or the third party), a brief introduction to each case, court verdict if available, announcement date, etc. From this

database, I obtain a sample of various types of litigation for 3,301 lawsuits over the period of 1999-2011 with either defendant or plaintiff listed on the Shanghai and Shenzhen Stock Exchanges. In total, each of the litigant samples (defendant and plaintiff) comprises 10,489 firm year observations. I obtain forced CEO turnover data from China Stock Market and Accounting Research (GTA/CSMAR), and in total, I identify 1512 forced CEO turnovers over the same period (1999-2011). The data on firm characteristics, operating performance, and the corporate governance indicators are also taken from CCER/Sinofin. The state ownership data and the corporate governance variables are from GTA/CSMAR.

4.2 Models

To test the hypotheses on determinants of corporate litigation (H1 and H2), I apply the following probit regression model in both of the defendant and plaintiff samples respectively:

$$Litigation_t = a_0 + a_1 Analyst_t + a_2 SOE_t + \sum_{i=3}^n a_i Control_t + Year + Industry + \varepsilon$$

(1)

where *litigation* is the dependant variable, which represents defendant, plaintiff, and the verdict results such as win and not win. When *litigation* represents defendant, for example, this equation aims to find out the effect of being an SOE and the impact of information asymmetry on the probability of a firm being sued. The same logic applies when litigation represents plaintiff. In addition, when *litigation* represents defendant winning the case (*Defendant_Win*), I try to find the effects of being an SOE on the probability of the defendant winning the lawsuit. The same logic applies when litigation represents other court verdicts.

Analyst is the number of analysts following the firm, which is a proxy I use for information asymmetry (Hong, Lim and Stein, 2000; Clarke and Shastri, 2000; Chae, 2005; Uysal, Kedia,

and Panchapagesan, 2008). As argued by Chae (2005), since analysts pose their forecasts of certain firms in various public and private media channels, individual investors can obtain this information more easily if there are more analysts covering the firm, thus mitigating information asymmetry. Hong, Lim and Stein (2000) also state that holding all else equal, the more analysts covering the company, the more firm-specific information will be generated and the faster that information will be obtained by the public, leading to a lower level of information asymmetry.

The dependant variable *SOE* is a dummy variable equal to 1 if the firm is state-owned and 0 otherwise. The control variables cover firm performance, firm characteristics and governance characteristics, including industry-median adjusted return on assets (*ROA*), dummy variable of financial distress (*ST* is equal to 1 if the firm is in financial distress, and 0 otherwise), the natural logarithm of total market capitalisation (*SIZEM*), firm's total sales growth rate (*GRWOTH*), firm's budget on research and development (*RD*), the reciprocal of proportion of firm's intangible assets (*TANGIBLE*)(Peng and Röell, 2008), firm's leverage ratio measured as total debt over total assets (*LEVERAGE*), dummy variable of duality (*DUAL* is equal to 1 if CEO holds the position of board chair, and 0 otherwise), board independence as measured by the proportion of outside directors(*BIND*), board size (*BOARDSIZE*) and supervisory board size (*SUPERVISORYSIZE*) measured by the number of people in the board and supervisory board respectively, and board meeting frequency (*BMEETF*) and supervisory board meeting frequency (*SMEETF*) measured by the number of meetings held by these two boards per year respectively.

Year and industry dummies are included to control for the time and industry effects. Industry dummy variables are constructed based on the first two digits of the Global Industry Classification Standard (GICS) codes.

In Equation (1), variables *SOE* and *Analyst* capture the effects of being a SOE and information asymmetry on the probability of litigation. If the coefficient $a_1 < 0$ in both samples, then the high analyst coverage that mimics low information asymmetry is associated with a reduction in the probability of corporate litigation, supporting my prediction in H1. If the coefficient $a_2 < 0$ ($a_2 > 0$) in the defendant (plaintiff) sample, then being a SOE is associated with lower (higher) probability of being sued (suing), supporting my prediction in H2.

To test the main hypotheses, H3a, b and H4, I apply the following probit regression model in each of the litigant samples, respectively:

$$\begin{aligned} \text{Forced Ceoturnover}_{t+1} = & a_0 + a_1 \text{Litigation}_t + a_2 \text{Positive Court verdict}_t + \\ & a_3 \text{Negative Court verdict}_t + \sum_{i=3}^n a_i \text{Control}_t + \text{Year} + \\ & \text{Industry} + \varepsilon \end{aligned} \quad (2)$$

ForcedCEOturnover, the one-year lead dependant variable, assigned to 1 if the CEO is forced to leave and 0 otherwise. A CEO departure is identified as forced turnover if the database labelled the reason for leave as one of the followings: 1. unexpected resignation; 2. fired; 3. leave for health reasons; 4. leave for personal reasons; 5. leave for legal charges; 6. others. The forced CEO turnover doesn't include natural retirement. *Litigation* represents firm being sued (*Defendant*) when I apply this model in the defendant sample and firm suing other companies (*Plaintiff*) in the plaintiff sample. Court verdict represents winning or not winning the case in both litigants' samples. The same set of control variables are used to control for firm characteristics. I add the dividend-adjusted individual share return (*RET*) over the fiscal year and the natural logarithm of firm book value at the end of the fiscal year (*SIZEB*) as extra controls on firm's characteristics. Year and industry dummies are also incorporated to control for the year and industry effects.

In Equation 2, if $a_1 > 0$ in both samples, then being involved in litigation, either as defendant or plaintiff, is associated with higher probability of forced CEO turnover, supporting the main prediction in H3a that CEOs are forced by board of directors to assume responsibility in the event of corporate litigation. To lend support to H4, $a_2 < 0$ and $a_3 > 0$ need to hold in both the litigants' samples, indicating that positive (negative) court verdict is associated with decrease (increase) of the probability of forced CEO turnover.

4.3 Sample Selection Bias and Propensity Score Matching

The effects of litigation on CEO turnover may be subject to sample selection bias in a sense that corporate litigation and CEO turnover may be affected by the same factors, such as firm performance, financial distress, firm size, board characteristics, etc. I therefore apply the propensity-score matching (PSM) method to resolve this endogeneity issue (Rosenbaum and Rubin, 1985). PSM is a statistical matching procedure that predicts the effects of a treatment (litigation) by accounting for the factors that estimate the receipt of such treatment (Hou, 2014; Rosenbaum and Rubin, 1985). To address the potential endogeneity issue, I would hope to compare the CEO turnover of a firm that is involved in corporate litigation with an otherwise identical firm that is not involved in lawsuit. PSM helps me to find such matching firms based on the propensity-scores calculated by accounting for the factors that lead to the treatment factor (litigation), so to provide me a less biased setting for this study.

To obtain the matching sample, I first apply a probit regression model on the treatment factor (litigation in this case) based on the firm characteristics, firm performance and governance features, including *SOE*, *ROA*, *ST*, *RET*, *SIZE*, *SHARETURNOVER*, *LEVERAGE*, *TANGIBLE*, *DUALITY*, *BIND*, *CommissionNumber*, *Bsize* and *Sbsize*. *BIND* is a proxy of board independence calculated as the proportion of independence directors on board.

CommissionNumber, a variable that I use to control for governance, represents the number of commission that a firm has established to specifically improve the governance. I then construct a one-to-one nearest-neighbour matched sample based on the propensity-score from the above probit regression. This PSM matched sample is less biased by the endogeneity issue, because the treatment group (litigated group) and the control group (not litigated) are near identical except the treatment factor. I then apply the following model to this matched sample.

$$\text{Forced Ceoturnover}_{t+1} = a_0 + a_1 \text{Litigation}_t + a_2 \text{Positive Court verdict}_t + a_3 \text{Negative Court verdict}_t + \sum_{i=3}^n a_i \text{Control}_t + \varepsilon \quad (3)$$

where all the variables used here are same as in Equation (2). For the results to support the hypothesis H3a, I need the coefficients $a_1 > 0$ in both defendants and plaintiffs samples, indicating that being involved in litigation, either as defendant or plaintiff, is associated with higher probability of forced CEO turnover. To lend support to H4, $a_2 < 0$ and $a_3 > 0$ need to hold in both the litigants' samples, indicating that positive (negative) court verdict is associated with decrease (increase) of the probability of forced CEO turnover.

4.4 Additional Tests

During the economic reform, the abolishment of planned economy was not sudden but gradual. Consequently within the Chinese state sector, legal means of supervision and control have not completely replaced the administrative directives. Even today, the leadership of some state-owned enterprises (SOEs), is still selected and nominated by the Communist Party (Clarke et al., 2007). This entrenchment of managers provided by the state thus has the power to undermine the effectiveness of corporate governance. I therefore use *SOE*, representing state ownership, as a proxy for governance. Fama and Jensen (1983) state that outside directors are more efficient than inside directors in monitoring management because they signal their

abilities in the labour market through board decisions. Consistent with Fama and Jensen (1983), Weisbach (1988) provide empirical evidence that outside directors are more likely than insider directors to replace poorly performing CEOs. If outside directors are associated with a higher probability of replacing unfit CEOs, then the more independent board, as measured by a higher proportion of outside directors in board, should be expected to be associated with higher levels of forced CEO replacement. I therefore use *BIND*, representing board independence, as another proxy for corporate governance. To test the predictions made in H5 and H6, I apply the following probit models:

$$\begin{aligned}
\text{Forced Ceoturnover}_{t+1} = & a_0 + a_1 \text{Litigation}_t + a_2 \text{Positive Court verdict}_t + \\
& a_3 \text{Negative Court verdict}_t + a_1 \text{Litigation}_t \times \text{SOE} + \\
& a_2 \text{Positive Court verdict}_t \times \text{SOE} + \\
& a_3 \text{Negative Court verdict}_t \times \text{SOE} + \\
& \sum_{i=3}^n a_i \text{Control}_t + \text{Year} + \text{Industry} + \varepsilon
\end{aligned}$$

$$\begin{aligned}
\text{Forced Ceoturnover}_{t+1} = & a_0 + a_1 \text{Litigation}_t + a_2 \text{Positive Court verdict}_t + \\
& a_3 \text{Negative Court verdict}_t + a_1 \text{Litigation}_t \times \text{BIND} + \\
& a_2 \text{Positive Court verdict}_t \times \text{BIND} + \\
& a_3 \text{Negative Court verdict}_t \times \text{BIND} + \\
& \sum_{i=3}^n a_i \text{Control}_t + \text{Year} + \text{Industry} + \varepsilon
\end{aligned}$$

$$\begin{aligned}
\text{Forced Ceoturnover}_{t+1} = & a_0 + a_1 \text{Litigation}_t + a_2 \text{Positive Court verdict}_t + \\
& a_3 \text{Negative Court verdict}_t + a_1 \text{Litigation}_t \times \text{ST} + \\
& a_2 \text{Positive Court verdict}_t \times \text{ST} + \\
& a_3 \text{Negative Court verdict}_t \times \text{ST} + \\
& \sum_{i=3}^n a_i \text{Control}_t + \text{Year} + \text{Industry} + \varepsilon \tag{4}
\end{aligned}$$

where all variables are the same as in Equation (3) except the newly added interaction variables used to capture the entrenchment effects of state ownership, the effects of board independence and the effects of financial distress. To support H5, I expect the coefficients of variables interacted with *SOE* to be negative, indicating the CEOs of SOE are entrenched by the state ownership, and the coefficient on variables interacted with *BIND* to be positive, indicating that board independence increases the board effectiveness to discipline unqualified CEOs. I also expect the coefficients of variables interacted with *ST* to be positive, showing that the litigation indeed causes more damage to financially distressed firms and thus the probability of the CEO being forced to leave increases.

5. Empirical Results

5.1 Descriptive Statistics

In Table 1, I provide descriptive statistics for corporate litigations by type filed from 1999 to 2011. Each of these cases involves a listed firm either as defendant or plaintiff. There are a large range of litigation types according to the CCER/Sinofin database. By reading the description attached to each case, I classify each case according to what appears to be the real issue that caused the filing of the lawsuit. Breach of contract includes a wide spectrum of commercial contract breaches, such as delivery, pricing, quality and ownership of goods and services. Breach of loan covenants typically involves one corporate seeking repayment from another. In some cases, banks are also involved in this type of breach as plaintiffs. Breach of loan guarantee agreement arises when a firm agrees to cover another firm's debt but then fails to do so. When the company that has originally taken the loan is not able to repay the principal or interest, the debt holder, typically a bank, seeks legal resolution by filing a lawsuit against the borrower and its guarantees. Table 1 shows that these three types of lawsuits are the most

common ones in China, accounting for about 85% of lawsuits filed. For breach of loan covenants and breach of loan guarantee agreement, the numbers of cases are both significant in 2004 and 2005. This is because the Chinese government issued a requirement that led the banks to urgently improve their balance sheets. As a result, banks resorted to legal measures in order to seek repayment of loans (Firth et al., 2011). Disputes over financial service typically involve the service providers being accused of inappropriately managing the funds on which they have taken fiduciary duties and disputes over intangibles include theft of intellectual property rights, copyright, patent infringements and non-payment of royalties due.

In Table 2, I present the descriptive statistics of the variables in the firm-year samples in panel A. I also provide the PSM matched samples in Panel B, C, E and F. Panel D and G provide differences of medians of each variable from the treatment group (litigated firms) and the PSM matched control group (non-litigated firms). It shows that there are no statistically significant difference between the treatment and control group except the treatment factor (litigation), indicating that the PSM provides a near perfect matching sample. This allows me to conduct the tests in a close-to-unbiased environment. Table 3 presents the correlation matrix of the variables in the firm-year samples. *ForcedCEOturnover* is significantly correlated with both *Plaintiff* and *Defendant*. In the subsequent empirical analysis, the number of observations sometimes vary from those listed in Table 2 because of missing data and observations automatically dropped in regression.

[Insert Table 1, 2, and 3 about here]

5.2 Test of Hypotheses 1 and 2

To test the hypotheses about the impact of information asymmetry and effects of SOE on corporate litigation, I regress lawsuit (*Defendant* and *Plaintiff*) and court verdict

(*Defendant_Win*, *Defendant_NotWin*, *Plaintiff_Win*, and *Plaintiff_NotWin*) on the proxies of information asymmetry (*Analyst*) and SOE. Results are presented in Table 4. The coefficient of *Analyst* in both defendant and plaintiff samples are significant and negative to the controls of firm characteristics, performance and board characteristics, supporting H1 that firms with high analyst coverage, which mimics low information asymmetry, are less likely to be involved in corporate litigations. For SOE, the coefficient is significant and negative in the defendant sample. This indicates that being an SOE reduces the probability of being sued by other firms, lending support for H2. However, in the plaintiff sample, the coefficient of SOE, though positive, is not statistically significant. The coefficient of SOE is statistically significant and positive in the plaintiff's positive verdict model, showing that SOEs are more likely to win their cases than other firms when they initiate the lawsuits. The results are in line with previous studies stating that SOEs are indeed favoured in the judicial system (Potter, 1999; MacNeil, 2002; Clarke et al., 2006; Firth et al., 2011). My empirical test results on information asymmetry are also consistent with theories (Bebchuk, 1984; Png, 1987; Reinganum and Wilde, 1986; Schweitzer, 1989).

[Insert Table 4 about here]

5.3 Test for Main Hypotheses 3, 4, 5 and 6

To test the main hypotheses 3, 4, 5 and 6, I regress ForcedCEOturnover on litigation (*Defendant* and *Plaintiff*) and court verdicts (*Defendant_Win*, *Defendant_NotWin*, *Plaintiff_Win*, and *Plaintiff_NotWin*) in both the original and the PSM matched samples. Table 5 presents the results. In Panel A, Model 1 shows that the coefficient on *Defendant* is marginally significant and positive, indicating that CEOs are indeed held accountable for corporate litigations. In Model 2, however, when I put both defendant and court verdicts in the regression, none of the coefficients on them are significant. I suspect that this may be due to

the endogeneity issue aforementioned, e.g., the factors that cause the corporate litigation may also cause the forced CEO turnover. To resolve this issue, I apply both Model 1 and Model 2 in the PSM matched sample, which is constructed to mitigate the endogeneity problem. The results are presented in Panel B. This time, the coefficients of *Defendant* in both models are statistically significant, indicating that CEOs are indeed held accountable for lawsuits. The results here support the main hypothesis H3a. In Panel C and D, I regress the forced CEO turnover on plaintiff and court verdicts in both the original sample and the PSM sample. The coefficients of *Plaintiff* are statistically significant in the PSM sample, once again supporting the prediction in H3a. From the results in Panel C and D, I can also see that the court verdict plays an important role in whether the CEO stays or leaves. Specifically, the coefficient of plaintiff winning the case (*Plaintiff_Win*) is significant at the 1% level, indicating that when the plaintiff wins a lawsuit, the probability of forced CEO turnover is reduced significantly. This supports the hypothesis H4. I do not observe the same effects for defendant court verdicts.

Table 5 also presents results that are consistent with the predictions in hypotheses H5, and H6 that the sensitivities of CEO turnover and corporate litigation are enhanced in firms with effective corporate governance and when firms are in financial distress. To illustrate, the coefficients on SOE are significantly negative at the 1% level in both Panel A and Panel C⁶, indicating that CEOs in SOEs are entrenched, protected by the state ownership. Because typically the senior management team in SOEs are selected and nominated directly by the government in China, it is difficult for the board of directors to force them to leave even though they are not representing the best interest of their shareholders (Clarke et al., 2007). For financially distressed firms, I am expecting that CEOs are more likely to be replaced due to

⁶ The results presented in Panel B and D are not used as evidence of H6 and H7 because the firms in the PSM sample are matched by the variables including SOE, ST and IND in order to provide an ideal setting to test litigation effects on CEO turnover. So it is obvious that coefficients of SOE and ST are not significant in that sample.

the more severe damage that litigation may cause to such firms (Cutler and Summers, 1988; Bhagat, Brickley and Coles, 1994). Table 5 shows that the coefficients for ST are positive and significant, indicating that financial distress is indeed associated with higher probability of forced CEO turnover. Thus H5 and H6 are largely supported by the evidence.

[Insert Table 5 about here]

Table 6 provides the results of additional tests of the interaction effects between litigation and SOE, ST, and BIND respectively. Panel A presents the results for both the original and the PSM samples for defendant firms and Panel B presents the results for the plaintiff firms. None of the coefficients for the interactive terms in Panel A are significant at the 5% level. However in Panel B, the coefficients of state-owned plaintiff firm winning the case (*Plaintiff_Win* × *SOE*) are significantly negative, supporting the SOE entrenchment theory (H5, poor governance), i.e. CEOs are less likely to be replaced in state-owned firms. Coefficients of plaintiff firm with a more independent board winning the lawsuit (*Plaintiff_Win* × *Ind*) are statistical significant and positive, supporting the prediction in H5 that board independence (good governance) is positively associated with disciplining CEOs who do not act in the best interest of shareholders. None of other coefficients of intersection terms is significant at the 5% significance level.

[Insert Table 6 about here]

6. Conclusion

It seems that a consensus has already been formed in the U.S. that corporate lawsuits exert negative effects on defendant firms' stock performance, reputation and business opportunities (Culter and Summers, 1988; Englemann and Cornell, 1988; Bhagat, Brickley, and Coles, 1994,

1998; Muoghalu, Robison, and Glascock, 1990; Hersch, 1991; Koku, Qureshi, and Akhigbe, 2001; Cumming, Haslem and Knill, 2011). Study by Firth et al. (2011) extends these results to China and documents that both the plaintiff and defendant firms are suffering from negative abnormal share performance due to corporate lawsuits. However, little research has been done to examine the consequence of corporate lawsuits to corporate governance. This paper fills this research gap by examining whether CEOs are taking responsibility for corporate litigation. I find, after controlling for possible selection bias and various firm characteristics, that both firms in litigations see a subsequent higher probability of forced CEO turnover. Further, I document that plaintiffs' positive court verdict also affects the probability of CEOs being forced to leave. Specifically, if the plaintiff wins a lawsuit, the probability of the CEO being fired is significantly reduced.

The setting also allows me to test the effects of state ownership, financial distress and board independence on CEO turnover of firms being involved in corporate litigation. The results suggest that CEOs in firms with relatively poor governance, such as state-owned enterprises, are less likely to be forced to leave the firm even though they are not representing the best interest of shareholders. The evidence also supports that CEOs in firms with financial distress are more likely to be forced to leave, possibly as litigation may cause more damage to firms in a bad financial state. In addition, the results add to the evidence that corporate litigation can serve as an effective corporate governance mechanism. This paper also empirically tests the effects of information asymmetry on corporate litigation. The evidence supports the theory that lower information asymmetry is associated with lower probability of lawsuits.

This paper also offers important practical implications. For corporate governance, CEO turnover and its link to corporate lawsuits can provide information on how effectively Chinese listed firms solve one of the main principal-agent problems: the diverging interests between manager and shareholders. In order to help address the agency problems, it also provides

practical guidance on further improving the effectiveness of corporate internal monitoring mechanism, for example that state-owned enterprises need to be further privatised and the selection and nomination process of senior management team in SOEs should be independent from the state. In addition, the results also provide some inputs for the deliberations of senior management as they consider improvements in representing the best interest of shareholders. Since China is an increasingly influential emerging country, this paper also has policy implications for other transitional economies.

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Appendix A: Variable Definitions

Variables	Definitions
<i>Defendant</i>	Dummy variable equal to 1 if the firm has been sued, and 0 otherwise.
<i>Defendant_Win</i>	Dummy variable equal to 1 if the firm has been sued and won the case, and 0 otherwise.
<i>Defendant_NotWin</i>	Dummy variable equal to 1 if the firm has been sued and not won the case, and 0 otherwise.
<i>Plaintiff</i>	Dummy variable equal to 1 if the firm has sued other firm, and 0 otherwise.
<i>Plaintiff_Win</i>	Dummy variable equal to 1 if the firm has sued other firm and won the case, and 0 otherwise.
<i>Plaintiff_NotWin</i>	Dummy variable equal to 1 if the firm has sued other firm and not won the case, and 0 otherwise.
<i>Breach of Contract</i>	Dummy variable equal to 1 if the firm is involved in lawsuit of breach of contract, and 0 otherwise.
<i>Breach of Loan Co.</i>	Dummy variable equal to 1 if the firm is involved in lawsuit of breach of loan covenants, and 0 otherwise.
<i>Breach of Loan Gu.</i>	Dummy variable equal to 1 if the firm is involved in lawsuit of breach of loan guarantee agreement, and 0 otherwise.
<i>Financial Service D.</i>	Dummy variable equal to 1 if the firm is involved in lawsuit of financial service disputes, and 0 otherwise.
<i>Disputes over Inta.</i>	Dummy variable equal to 1 if the firm is involved in lawsuit of disputes over intangible assets, and 0 otherwise.
<i>CEOTurnvoer</i>	Dummy variable equal to 1 if the CEO has been forced to leave the firm, and 0 otherwise.
<i>Analyst</i>	The number of analyst that are covering the company.
<i>SOE</i>	Dummy variable equal to 1 for state-owned enterprises, and 0 otherwise.
<i>ROA</i>	Industry-median adjusted return on assets.
<i>ST</i>	Dummy variable equal to 1 if the firm is in financial distress, and 0 otherwise.
<i>RET</i>	Individual dividend-adjusted share return.
<i>SIZEM</i>	Firm's total market capitalization
<i>SIZEB</i>	Natural logarithm of firm book value at the end of year.
<i>LEVERAGE</i>	Leverage ratio calculated as total debt over total assets.
<i>RD</i>	Firm expenses on research and development.
<i>GROWTH</i>	Total sales growth rate calculated as total sales over previous year's total sales at the end of year.
<i>SHARETURNOVER</i>	Daily average share turnover of individual firm over the year.
<i>TANGIBLE</i>	Total tangible asset over firm's total assets at the end of the year.
<i>DUAL</i>	Dummy variable equal to 1 if CEO holds the position of the board chair and 0 otherwise.
<i>BIND</i>	The number of outside directors divided by the board size.
<i>CommisionNumber</i>	The number of commissions the firm has put in place to improve corporate governance.
<i>BDSIZE</i>	The total number of directors on board at the end of year.
<i>SBSIZE</i>	The total number of directors on supervisory board at the end of year.
<i>BMEETF</i>	The total number of directors' board meetings in one year.

<i>SMEETF</i>	The total number of supervisory board meetings in one year.
<i>Year Effect</i>	The year ranges from 1999 to 2011.
<i>Industry Effects</i>	The first two digits of the Global Industry Classification Standard (GICS) is utilized to construct the industry dummy variables to mimic the industry effects.

Appendix B: Number of Lawsuits

Table A Descriptive Statistics of Number of Lawsuits

From the year 2002 and onward, economics disputes and maritime law and affairs are included in civil disputes. Period covers from 1979, the outset of the market reform, to the year 2011, which consistent with sample in the empirical analysis. Data source is from National Statistics Bureau of China.

Year	Total Case Accepted	Criminal	Civil	Economics Disputes	Administrative	Maritime Law and Affairs
1979	513,789	123,846	389,943			
1980	763,535	197,856	565,679			
1981	906,051	232,125	673,926			
1982	1,024,160	245,219	778,941			
1983	1,343,164	542,648	756,436	43,553	527	
1984	1,355,460	431,357	838,307	84,813	983	
1985	1,319,741	246,655	846,391	225,541	916	238
1986	1,611,282	299,720	989,409	321,220	632	301
1987	1,875,229	289,614	1,213,219	366,110	5,940	346
1988	2,290,624	313,306	1,455,130	513,046	8,573	569
1989	2,913,515	392,564	1,815,385	694,907	9,934	725
1990	2,916,774	459,656	1,851,897	591,462	13,006	753
1991	2,901,685	427,840	1,880,635	566,592	25,667	951
1992	3,051,157	422,991	1,948,786	650,601	27,125	1,654
1993	3,414,845	403,267	2,089,257	892,580	27,911	1,830
1994	3,955,475	482,927	2,383,764	1,051,742	35,083	1,959
1995	4,545,676	495,741	2,718,533	1,275,959	52,596	2,847
1996	5,312,580	618,826	3,093,995	1,515,848	79,966	3,945
1997	5,288,379	436,894	3,277,572	1,478,822	90,557	4,534
1998	5,410,798	482,164	3,375,069	1,450,049	98,350	5,166
1999	5,692,434	540,008	3,519,244	1,529,877	97,569	5,736
2000	5,356,294	560,432	3,412,259	1,290,867	85,760	6,976
2001	5,344,934	628,996	3,459,025	1,149,101	100,921	6,891
2002	5,132,199	631,348	4,420,123		80,728	
2003	5,130,760	632,605	4,410,236		87,919	
2004	5,072,881	647,541	4,332,727		92,613	
2005	5,161,170	684,897	4,380,095		96,178	
2006	5,183,794	702,445	4,385,732		95,617	
2007	5,550,062	724,112	4,724,440		101,510	
2008	6,288,831	767,842	5,412,591		108,398	
2009	6,688,963	768,507	5,800,144		120,312	
2010	6,999,350	779,595	6,090,622		129,133	
2011	7,596,116	845,714	6,614,049		136,353	

Table 1 Descriptive Statistics of the Sample of Lawsuits Filed

Statistics are for corporate litigations in which at least one side, defendant or plaintiff is a listed firm in China. Cases are from CCER/Sinofin database. All cases here were filed from 1999 to 2011.

	N=3301 cases													
	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	Total
Type of case														
For All:														
Breach of contract	13	35	106	98	30	94	130	106	59	60	81	63	73	948
Breach of loan Covenants	8	87	279	201	56	192	264	197	113	104	117	35	31	1684
Breach of loan guarantee Agreement	6	20	55	27	11	27	14	13	19	4	7	14	8	225
Financial service disputes	1	0	6	8	7	43	25	25	7	11	6	0	4	143
Intangible	0	2	3	10	2	4	9	8	6	9	9	10	17	89
Other	3	6	21	33	7	13	28	16	14	21	22	11	17	212
Total	31	150	470	377	113	373	470	365	218	209	242	133	150	3301
For defendants:														
Breach of contract	9	16	64	45	14	42	65	66	28	33	56	43	40	521
Breach of loan Covenants	5	71	210	118	27	102	150	146	95	80	92	32	27	1155
Breach of loan guarantee Agreement	6	20	50	23	7	18	10	9	18	4	6	13	5	189
Financial service disputes	1	0	1	2	3	26	5	8	4	7	5	0	2	64
Intangible	0	2	1	4	1	2	3	2	5	5	3	4	12	44
Other	1	3	10	19	5	6	14	11	7	8	21	7	15	127
Total	22	112	336	211	57	196	247	242	157	137	183	99	101	2100
For plaintiffs:														
Breach of contract	4	19	42	53	16	52	65	40	31	27	25	20	33	427
Breach of loan Covenants	3	16	69	83	29	90	114	51	18	24	25	3	4	529
Breach of loan guarantee Agreement	0	0	5	4	4	9	4	4	1	0	1	1	3	36
Financial service disputes	0	0	5	6	4	17	20	17	3	4	1	0	2	79
Intangible	0	0	2	6	1	2	6	6	1	4	6	6	5	45
Other	2	3	11	14	2	7	14	5	7	13	1	4	2	85
Total	9	38	134	166	56	177	223	123	61	72	59	34	49	1201

Table 2 Descriptive Statistics

This table presents the descriptive statistics for the total sample, subsamples, and PSM samples. Plaintiff is assigned to 1 if the firm sues other companies, and 0 otherwise. Defendant is equal to 1 if the firm is being sued, and 0 otherwise. Other variables are defined in the appendix. The sample period covers 1999 to 2011. ***, **, * denote significance at 1%, 5%, and 10% levels, respectively.

Variable	Panel A				Panel B				Panel C				Panel D	
	Total				Defendant				PSM Matching Sample				Panel B - Panel C	
	Obs	Mean	Std. Dev.	Median	Obs	Mean	Std. Dev.	Median	Obs	Mean	Std. Dev.	Median	Mean Difference	
<i>Plaintiff</i>	10489	0.039	0.194	0										
<i>Plaintiff_Win</i>	10489	0.018	0.133	0										
<i>Plaintiff_NotWin</i>	10489	0.006	0.077	0										
<i>Defendant</i>	10489	0.034	0.181	0	328	1	0	1	328	0	0	0		
<i>Defendant_Win</i>	10489	0.005	0.069	0	328	0.116	0.321	0	328	0	0	0	-0.116	***
<i>Defendant_NotWin</i>	10489	0.018	0.131	0	328	0.491	0.501	0	328	0.009	0.095	0	-0.482	***
<i>CEO Turnover</i>	10489	0.070	0.256	0	328	0.137	0.345	0	328	0.058	0.234	0	-0.079	***
<i>SOE</i>	10489	0.687	0.464	1	328	0.610	0.489	1	328	0.652	0.477	1	0.043	
<i>BIND</i>	10489	0.307	0.123	0.333	328	0.300	0.124	0.333	328	0.303	0.130	0.333	0.004	
<i>ROA</i>	10489	0.003	0.061	0.002	328	-0.028	0.093	-0.011	328	-0.024	0.087	-0.013	0.003	
<i>ST</i>	10489	0.053	0.224	0	328	0.213	0.410	0	328	0.229	0.421	0	0.015	
<i>RET</i>	10489	0.130	0.603	0.004	328	0.170	0.641	0.024	328	0.183	0.676	0.011	0.013	
<i>SIZEM</i>	10489	21.866	1.104	21.750	328	21.417	1.157	21.348	328	21.511	1.048	21.557	0.094	
<i>SIZEB</i>	10489	20.613	1.136	20.529	328	19.750	1.247	19.746	328	19.814	1.161	19.822	0.064	
<i>LEVERAGE</i>	10489	0.490	0.188	0.495	328	0.581	0.197	0.608	328	0.593	0.197	0.619	0.012	
<i>TANGIBLE</i>	10489	0.289	0.184	0.262	328	0.267	0.193	0.223	328	0.268	0.175	0.250	0.001	
<i>DUAL</i>	10489	0.135	0.342	0	328	0.146	0.354	0	328	0.116	0.321	0	-0.030	
<i>BDSIZE</i>	10489	9.464	2.126	9	328	9.262	1.899	9	328	9.284	2.071	9	0.021	
<i>SBSIZE</i>	10489	4.136	1.408	3	328	4.027	1.228	3	328	4.006	1.334	3	-0.021	

Table 2 (Continued) Descriptive Statistics

This table presents the descriptive statistics for the total sample, subsamples, and PSM samples. Plaintiff is assigned to 1 if the firm sues other companies, and 0 otherwise. Defendant is equal to 1 if the firm is being sued, and 0 otherwise. Other variables are defined in the appendix. The sample period covers 1999 to 2011. ***, **, * denote significance at 1%, 5%, and 10% levels, respectively.

Variable	Panel E				Panel F				Panel G	
	Plaintiff				PSM Matching Sample				Panel E - Panel F	
	Obs	Mean	Std. Dev.	Median	Obs	Mean	Std. Dev.	Median	Mean Difference	
<i>Plaintiff</i>	376	1	0	1	376	0	0	0		
<i>Plaintiff_Win</i>	376	0.420	0.494	0	376	0.003	0.052	0	-0.418	***
<i>Plaintiff_NotWin</i>	376	0.141	0.348	0	376	0	0	0	-0.141	***
<i>CEO Turnover</i>	376	0.109	0.312	0	376	0.061	0.240	0	-0.048	
<i>SOE</i>	376	0.694	0.461	1	376	0.707	0.456	1	0.013	
<i>BIND</i>	376	0.301	0.121	0.333	376	0.295	0.128	0.333	-0.006	
<i>ROA</i>	376	-0.016	0.073	-0.006	376	-0.015	0.074	-0.005	0.001	
<i>ST</i>	376	0.120	0.325	0	376	0.122	0.328	0	0.003	
<i>RET</i>	376	0.076	0.575	-0.041	376	0.016	0.516	-0.043	-0.060	
<i>SIZEM</i>	376	21.415	0.970	21.368	376	21.373	0.951	21.364	-0.042	
<i>SIZEB</i>	376	20.163	1.088	20.180	376	20.140	1.135	20.231	-0.023	
<i>LEVERAGE</i>	376	0.542	0.188	0.570	376	0.540	0.193	0.552	-0.003	
<i>TANGIBLE</i>	376	0.283	0.182	0.265	376	0.274	0.183	0.245	-0.009	
<i>DUAL</i>	376	0.128	0.334	0	376	0.120	0.325	0	-0.008	
<i>BDSIZE</i>	376	9.513	2.045	9	376	9.460	2.279	9	-0.053	
<i>SBSIZE</i>	376	4.261	1.555	4	376	4.298	1.422	5	0.037	

Table 3 Correlation Matrix

This table presents the correlation matrix for the variables in both the firm-year samples. Sample significance at 1% is indicated with in **bold** font.

		[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]	[17]	[18]
[1]	<i>Defendant</i>	1.000																	
[2]	<i>Defendant_Win</i>	0.324	1.000																
[3]	<i>Defendant_NotWin</i>	0.647	0.022	1.000															
[4]	<i>Plaintiff</i>	0.038	0.029	0.037	1.000														
[5]	<i>Plaintiff_Win</i>	0.030	0.074	0.015	0.621	1.000													
[6]	<i>Plaintiff_NotWin</i>	0.020	0.013	0.009	0.351	0.008	1.000												
[7]	<i>CEO Turnover</i>	0.046	0.013	0.037	0.025	0.012	0.013	1.000											
[8]	<i>SOE</i>	0.032	0.010	0.013	0.004	0.028	0.023	0.037	1.000										
[9]	<i>BIND</i>	0.005	0.009	0.015	0.003	0.000	0.004	0.051	0.164	1.000									
[10]	<i>ROA</i>	0.096	0.010	0.053	0.067	0.031	0.037	0.090	0.048	0.039	1.000								
[11]	<i>ST</i>	0.148	0.033	0.133	0.071	0.032	0.059	0.059	0.076	0.034	0.142	1.000							
[12]	<i>RET</i>	0.005	0.003	0.010	0.018	0.025	0.007	0.009	0.043	0.063	0.160	0.031	1.000						
[13]	<i>SIZEM</i>	0.083	0.038	0.060	0.090	0.060	0.047	0.054	0.118	0.057	0.316	0.177	0.280	1.000					
[14]	<i>SIZEB</i>	0.147	0.039	0.117	0.084	0.043	0.060	0.081	0.184	0.147	0.287	0.306	0.031	0.808	1.000				
[15]	<i>LEVERAGE</i>	0.090	0.018	0.088	0.050	0.014	0.040	0.013	0.043	0.111	0.312	0.148	0.038	0.013	0.086	1.000			
[16]	<i>TANGIBLE</i>	0.023	0.015	0.015	0.005	0.001	0.009	0.010	0.157	0.018	0.038	0.003	0.055	0.006	0.100	0.044	1.000		
[17]	<i>DUAL</i>	0.003	0.007	0.009	0.003	0.010	0.001	0.042	0.157	0.011	0.022	0.043	0.012	0.043	0.087	0.057	0.045	1.000	
[18]	<i>BDSIZE</i>	0.024	0.006	0.016	0.005	0.004	0.013	0.021	0.178	0.111	0.005	0.065	0.029	0.170	0.226	0.090	0.103	0.084	1.000
[19]	<i>SBSIZE</i>	0.014	0.001	0.005	0.017	0.018	0.018	0.015	0.246	0.118	0.009	0.061	0.011	0.149	0.191	0.064	0.131	0.074	0.368

Table 4 Determinants of Lawsuits

This table reports the probit regression results of the determinants of litigation. The dependant variables are Defendant, Defendant_Win, Defendant_NotWin, Plaintiff, Plaintiff_Win and Plaintiff_NotWin. Panel A covers the sample of defendant and Panel B covers the sample of plaintiff. The sample period is from 1999 to 2011. All the variables are defined in the appendix. Z-values are in parentheses ***, **, * denote significance at 1%, 5%, and 10% levels, respectively. Marginal effects are reported.

Variable	Panel A			Panel B		
	<i>Defendant</i>	<i>Defendant_Win</i>	<i>Defendant_NotWin</i>	<i>Plaintiff</i>	<i>Plaintiff_Win</i>	<i>Plaintiff_NotWin</i>
<i>SOE</i>	-0.007** (-1.97)	-0.001 (-1.13)	0.000 (0.20)	0.005 (1.34)	0.008*** (4.10)	-0.001 (-1.23)
<i>Analyst</i>	-0.006*** (-2.64)	-0.001 (-1.03)	-0.003** (-2.35)	-0.010*** (-3.93)	-0.003* (-1.84)	-0.002** (-1.99)
<i>ST</i>	0.062*** (6.01)	-0.001 (-0.83)	0.043*** (5.13)	0.025*** (3.01)	0.010* (1.93)	0.006* (1.94)
<i>ROA</i>	-0.068*** (-3.39)	0.005 (0.80)	-0.024** (-2.09)	-0.048** (-1.96)	-0.021 (-1.26)	-0.005 (-0.79)
<i>SIZEM</i>	-0.011*** (-5.23)	-0.002*** (-2.80)	-0.005*** (-3.78)	-0.003 (-1.31)	-0.003* (-1.90)	-0.000 (-0.24)
<i>GROWTH</i>	0.001 (0.60)	0.001 (1.31)	-0.000 (-0.22)	0.000 (0.18)	-0.000 (-0.01)	-0.001 (-0.85)
<i>RD</i>	0.000 (0.39)	-0.000 (-0.15)	0.000 (0.46)	-0.000 (-0.75)	0.000 (0.91)	-0.000 (-0.82)
<i>TANGIBLE</i>	-0.005 (-0.55)	0.006** (2.49)	-0.004 (-0.59)	0.001 (0.08)	0.003 (0.44)	-0.002 (-0.59)
<i>LEVERAGE</i>	0.049*** (8.64)	0.000 (0.21)	0.027*** (7.41)	-0.001 (-0.17)	0.000 (0.01)	0.002 (1.00)
<i>DUAL</i>	0.004 (1.01)	-0.001 (-1.56)	0.003 (1.15)	0.003 (0.64)	0.001 (0.25)	0.001 (0.49)
<i>BMEETF</i>	0.001** (2.29)	0.000*** (3.26)	0.001*** (2.80)	0.004*** (7.14)	0.001*** (3.27)	0.000 (1.23)
<i>SMEETF</i>	0.001 (1.05)	0.000 (1.36)	-0.000 (-0.26)	0.001 (1.18)	0.002*** (3.74)	0.001* (1.82)
<i>BDSIZE</i>	-0.000 (-0.09)	0.000 (0.09)	-0.000 (-0.28)	-0.000 (-0.10)	0.000 (0.83)	-0.000 (-0.29)
<i>SBSIZE</i>	-0.001 (-0.45)	-0.000 (-0.45)	-0.000 (-0.27)	0.002 (1.41)	0.000 (0.34)	-0.000 (-1.01)
<i>BIND</i>	0.017 (0.72)	-0.006 (-0.95)	0.012 (0.85)	-0.007 (-0.26)	-0.003 (-0.21)	-0.00 (-0.58)
<i>Industry effect</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year effect</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Pseudo R²</i>	0.171	0.100	0.201	0.084	0.105	0.093
<i>No. of Observations</i>	12876	12847	12876	12876	12847	12498

Table 5 The Litigation Effects on CEO_Turnover

This table presents the probit results of litigation on forced CEO turnover in both all firm-year sample and PSM matching sample of defendant. CEO_Turnover is a one-year lead dummy variable and it is assigned to 1 if CEO is forced to leave the firm, and 0 otherwise. Panel A covers the original sample of defendant and Panel B covers the PSM sample of defendant. The sample period is from 1999 to 2011. Other variables are defined in the appendix. Z-values are in parentheses. ***, **, * denote significance at 1%, 5%, and 10% levels, respectively. Marginal effects are reported.

Variable	Panel A		Panel B	
	All Sample		PSM Matching Sample	
	CEO_Turnover		CEO_Turnover	
	Model 1	Model 2	Model 1	Model 2
<i>Defendant</i>	0.027* (1.79)	0.013 (0.70)	0.067*** (3.19)	0.054** (1.99)
<i>Defendant_Win</i>		0.022 (0.55)		0.020 (0.40)
<i>Defendant_NotWin</i>		0.021 (0.83)		0.023 (0.75)
<i>SOE</i>	-0.018*** (-3.10)	-0.018*** (-3.10)	-0.066** (-2.47)	-0.066** (-2.47)
<i>IND</i>	-0.013 (-0.29)	-0.012 (-0.28)	-0.013 (-0.16)	-0.006 (-0.07)
<i>ROA</i>	-0.261*** (-6.53)	-0.263*** (-6.57)	-0.336*** (-2.93)	-0.346*** (-2.96)
<i>ST</i>	0.031** (2.34)	0.031** (2.32)	-0.004 (-0.14)	-0.004 (-0.17)
<i>RET</i>	-0.000 (-0.05)	-0.000 (-0.07)	0.005 (0.29)	0.005 (0.26)
<i>SIZEM</i>	0.003 (0.48)	0.003 (0.49)	-0.021 (-1.48)	-0.020 (-1.44)
<i>SIZEB</i>	-0.009* (-1.77)	-0.009* (-1.78)	0.005 (0.36)	0.005 (0.36)
<i>LEVERAGE</i>	-0.017 (-1.21)	-0.018 (-1.24)	-0.005 (-0.09)	-0.007 (-0.11)
<i>TANGIBLE</i>	-0.017 (-1.11)	-0.017 (-1.14)	-0.067 (-1.20)	-0.070 (-1.28)
<i>DUALITY</i>	0.023*** (2.91)	0.023*** (2.91)	0.009 (0.30)	0.009 (0.30)
<i>BDSIZE</i>	-0.001 (-0.70)	-0.001 (-0.70)	-0.009 (-1.43)	-0.010 (-1.43)
<i>SBSIZE</i>	-0.001 (-0.26)	-0.001 (-0.27)	0.011 (1.34)	0.012 (1.36)
<i>Industry effect</i>	Yes	Yes	No	No
<i>Year effect</i>	Yes	Yes	No	No
<i>Pseudo R²</i>	0.038	0.038	0.103	0.105
<i>No. of Observations</i>	10483	10483	656	656

Table 5 (Continued) The Litigation Effects on CEO_Turnover

This table presents the probit results of litigation on forced CEO turnover in both all firm-year sample and PSM matching sample of plaintiff. CEO_Turnover is a one-year leading dummy variable and it is assigned to 1 if CEO is forced to leave the firm, and 0 otherwise. Panel C covers the original sample of plaintiff and Panel D covers the PSM sample of plaintiff. The sample period is from 1999 to 2011. Other variables are defined in the appendix. Z-values are in parentheses. ***, **, * denote significance at 1%, 5%, and 10% levels, respectively. Marginal effects are reported.

Variable	Panel C		Panel D	
	All Sample		PSM Matching Sample	
	CEO_Turnover		CEO_Turnover	
	Model 1	Model 2	Model 1	Model 2
<i>Plaintiff</i>	0.019 (1.38)	0.055** (2.47)	0.046** (2.42)	0.073*** (3.25)
<i>Plaintiff_Win</i>		-0.046*** (-5.13)		-0.060*** (-3.40)
<i>Plaintiff_NotWin</i>		-0.022 (-1.01)		-0.021 (-0.79)
<i>SOE</i>	-0.019*** (-3.14)	-0.018*** (-3.06)	-0.039* (-1.65)	-0.031 (-1.34)
<i>IND</i>	-0.013 (-0.29)	-0.011 (-0.26)	-0.102 (-1.24)	-0.090 (-1.15)
<i>ROA</i>	-0.264*** (-6.59)	-0.263*** (-6.59)	-0.270** (-2.24)	-0.259** (-2.20)
<i>ST</i>	0.033** (2.44)	0.033** (2.44)	0.028 (0.81)	0.026 (0.77)
<i>RET</i>	-0.000 (-0.09)	-0.001 (-0.11)	0.025 (1.27)	0.022 (1.16)
<i>SIZEM</i>	0.003 (0.56)	0.003 (0.58)	-0.009 (-0.46)	-0.012 (-0.57)
<i>SIZEB</i>	-0.009* (-1.91)	-0.009* (-1.93)	-0.014 (-0.83)	-0.012 (-0.75)
<i>LEVERAGE</i>	-0.017 (-1.16)	-0.017 (-1.20)	-0.050 (-0.90)	-0.052 (-0.95)
<i>TANGIBLE</i>	-0.017 (-1.11)	-0.016 (-1.09)	-0.021 (-0.39)	-0.021 (-0.40)
<i>DUALITY</i>	0.023*** (2.92)	0.023*** (2.92)	0.011 (0.36)	0.010 (0.33)
<i>BDSIZE</i>	-0.001 (-0.73)	-0.001 (-0.75)	0.000 (0.07)	-0.000 (-0.02)
<i>SBSIZE</i>	-0.000 (-0.24)	-0.001 (-0.26)	0.000 (0.01)	-0.007 (-0.09)
<i>Industry effect</i>	Yes	Yes	No	No
<i>Year effect</i>	Yes	Yes	No	No
<i>Pseudo R²</i>	0.038	0.039	0.068	0.086
<i>No. of Observations</i>	10483	10483	752	752

TABLE 6 Interaction Effects on CEO_Turnover

This table presents the results of probit regression of forced CEO replacement. The dependant variable is CEO_Turnover in all models. CEO_Turnover is a one-year leading dummy variable assigned to 1 if CEO is forced to leave the firm and 0 otherwise. Panel A covers the sample of defendant and Panel B covers the sample of plaintiff. The sample period is from 1999 to 2011. Other variables are defined in the the appendix. Z-values are in parentheses. ***, **, * denote significance at 1%, 5%, and 10% levels, respectively. Marginal effects are reported.

Variable	Panel A					
	All Sample			PSM Matching Sample		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
<i>Defendant</i>	0.004 (0.16)	-0.015 (-0.41)	0.029 (1.27)	0.011 (0.30)	0.057 (0.82)	0.060* (1.95)
<i>Defendant_Win</i>	0.051 (0.65)	-0.064*** (-15.03)	0.018 (0.44)	0.054 (0.60)	-0.090*** (-4.43)	0.033 (0.55)
<i>Defendant_NotWin</i>	0.057 (1.12)	0.057 (0.75)	0.014 (0.53)	0.031 (0.70)	0.081 (0.90)	0.030 (0.85)
<i>Defendant×SOE</i>	0.015 (0.40)			0.095 (1.36)		
<i>Defendant_Win×SOE</i>	-0.029 (-0.68)			-0.038 (-0.75)		
<i>Defendant_NotWin×SOE</i>	-0.034 (-1.49)			-0.012 (-0.26)		
<i>Defendant×IND</i>		0.091 (0.68)			-0.010 (-0.05)	
<i>Defendant_Win×IND</i>		0.708* (1.82)			0.886 (1.40)	
<i>Defendant_NotWin×IND</i>		-0.086 (-0.58)			-0.168 (-0.86)	
<i>Defendant×ST</i>			-0.037* (-1.90)			-0.020 (-0.40)
<i>Defendant_Win×ST</i>			-0.000 (-0.00)			-0.052 (-1.32)
<i>Defendant_NotWin×ST</i>			0.023 (0.39)			-0.026 (-0.54)
<i>SOE</i>	-0.017*** (-2.89)	-0.018*** (-3.09)	-0.018*** (-3.12)	-0.118*** (-2.63)	-0.064** (-2.44)	-0.067** (-2.52)
<i>IND</i>	-0.012 (-0.28)	-0.020 (-0.45)	-0.013 (-0.30)	-0.017 (-0.20)	0.041 (0.34)	-0.010 (-0.12)
<i>ROA</i>	-0.263*** (-6.56)	-0.262*** (-6.55)	-0.265*** (-6.61)	-0.348*** (-3.08)	-0.332*** (-2.94)	-0.344*** (-2.97)
<i>ST</i>	0.030** (2.28)	0.031** (2.34)	0.038*** (2.59)	-0.004 (-0.17)	-0.001 (-0.02)	0.024 (0.54)
<i>RET</i>	-0.000 (-0.08)	-0.000 (-0.10)	-0.000 (-0.01)	0.004 (0.25)	0.003 (0.17)	0.008 (0.42)
<i>SIZEM</i>	0.003 (0.51)	0.003 (0.53)	0.002 (0.42)	-0.019 (-1.43)	-0.018 (-1.28)	-0.023 (-1.63)
<i>SIZEB</i>	-0.009* (-1.79)	-0.009* (-1.82)	-0.008* (-1.69)	0.005 (0.38)	0.002 (0.13)	0.006 (0.44)
<i>LEVERAGE</i>	-0.017 (-1.22)	-0.017 (-1.23)	-0.018 (-1.28)	-0.001 (-0.02)	-0.014 (-0.25)	-0.010 (-0.17)
<i>TANGIBLE</i>	-0.017 (-1.14)	-0.017 (-1.17)	-0.017 (-1.14)	-0.063 (-1.15)	-0.077 (-1.44)	-0.072 (-1.34)
<i>DUALITY</i>	0.023*** (2.93)	0.023*** (2.88)	0.023*** (2.92)	0.012 (0.39)	0.004 (0.15)	0.010 (0.32)
<i>BDSIZE</i>	-0.001 (-0.70)	-0.001 (-0.71)	-0.001 (-0.73)	-0.010 (-1.42)	-0.008 (-1.29)	-0.010 (-1.45)
<i>SBSIZE</i>	-0.001 (-0.28)	-0.001 (-0.26)	-0.000 (-0.24)	0.011 (1.31)	0.012 (1.47)	0.013 (1.49)
<i>Industry effect</i>	Yes	Yes	Yes	No	No	No
<i>Year effect</i>	Yes	Yes	Yes	No	No	No
<i>Pseudo R²</i>	0.039	0.039	0.039	0.112	0.114	0.108
<i>Obs</i>	10483	10483	10483	656	656	656

TABLE 6 (Continued) Interaction Effects on CEO_Turnover

This table presents the results of probit regression of forced CEO replacement. The dependant variable is CEO_Turnover in all models. CEO_Turnover is a one-year leading dummy variable assigned to 1 if CEO is forced to leave the firm and 0 otherwise. Panel A covers the sample of defendant and Panel B cover sample of plaintiff. The sample period is from 1999 to 2011. All other variables are defined in the appendix. Z-values are in parentheses. ***, **, * denote significance at 1%, 5%, and 10% levels, respectively. Marginal effects are reported.

Variable	Panel B					
	All Sample			PSM Matching Sample		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
<i>Plaintiff</i>	0.040 (1.25)	0.046 (0.90)	0.056** (2.28)	0.045 (1.34)	0.063 (1.21)	0.077*** (3.18)
<i>Plaintiff_Win</i>	0.012 (0.30)	-0.065*** (-17.55)	-0.048*** (-5.06)	0.025 (0.53)	-0.109*** (-4.74)	-0.059*** (-3.22)
<i>Plaintiff_NotWin</i>	-0.051*** (-3.04)	0.162 (0.91)	-0.020 (-0.77)	-0.053** (-2.18)	0.170 (0.98)	-0.015 (-0.48)
<i>Plaintiff×SOE</i>	0.018 (0.51)			0.038 (0.82)		
<i>Plaintiff_Win×SOE</i>	-0.061*** (-9.41)			-0.089*** (-4.94)		
<i>Plaintiff_NotWin×SOE</i>	0.154 (0.93)			0.137 (0.85)		
<i>Plaintiff×IND</i>		0.019 (0.17)			0.026 (0.16)	
<i>Plaintiff_Win×IND</i>		0.315*** (2.65)			0.328** (2.15)	
<i>Plaintiff_NotWin×IND</i>		-0.425* (-1.73)			-0.449* (-1.75)	
<i>Plaintiff×ST</i>			-0.002 (-0.06)			-0.021 (-0.46)
<i>Plaintiff_Win×ST</i>			0.025 (0.32)			-0.010 (-0.13)
<i>Plaintiff_NotWin×ST</i>			-0.010 (-0.16)			-0.018 (-0.29)
<i>SOE</i>	-0.017*** (-2.91)	-0.018*** (-3.09)	-0.018*** (-3.06)	-0.031 (-1.03)	-0.034 (-1.46)	-0.031 (-1.35)
<i>IND</i>	-0.010 (-0.22)	-0.010 (-0.24)	-0.011 (-0.26)	-0.071 (-0.98)	-0.082 (-0.77)	-0.093 (-1.20)
<i>ROA</i>	-0.259*** (-6.49)	-0.264*** (-6.60)	-0.263*** (-6.59)	-0.213* (-1.92)	-0.252** (-2.19)	-0.259** (-2.21)
<i>ST</i>	0.033** (2.46)	0.033** (2.43)	0.033** (2.32)	0.027 (0.85)	0.025 (0.74)	0.048 (0.90)
<i>RET</i>	-0.001 (-0.17)	-0.001 (-0.14)	-0.001 (-0.12)	0.018 (1.00)	0.019 (1.04)	0.022 (1.17)
<i>SIZEM</i>	0.003 (0.61)	0.004 (0.65)	0.003 (0.58)	-0.007 (-0.39)	-0.006 (-0.36)	-0.011 (-0.62)
<i>SIZEB</i>	-0.010** (-1.98)	-0.010** (-1.99)	-0.009* (-1.92)	-0.015 (-1.02)	-0.014 (-0.91)	-0.011 (-0.70)
<i>LEVERAGE</i>	-0.017 (-1.22)	-0.017 (-1.19)	-0.017 (-1.20)	-0.053 (-1.06)	-0.048 (-0.89)	-0.054 (-1.00)
<i>TANGIBLE</i>	-0.016 (-1.06)	-0.016 (-1.09)	-0.016 (-1.08)	-0.011 (-0.22)	-0.021 (-0.39)	-0.022 (-0.41)
<i>DUALITY</i>	0.023*** (2.94)	0.023*** (2.94)	0.023*** (2.92)	0.010 (0.36)	0.014 (0.47)	0.010 (0.32)
<i>BDSIZE</i>	-0.001 (-0.76)	-0.001 (-0.75)	-0.001 (-0.75)	-0.000 (-0.10)	-0.000 (-0.03)	0.000 (0.02)
<i>SBSIZE</i>	-0.001 (-0.26)	-0.000 (-0.25)	-0.001 (-0.27)	-0.001 (-0.10)	0.000 (0.01)	-0.000 (-0.06)
<i>Industry effect</i>	Yes	Yes	Yes	No	No	No
<i>Year effect</i>	Yes	Yes	Yes	No	No	No
<i>Pseudo R²</i>	0.041	0.041	0.039	0.115	0.099	0.087
<i>No. of Observations</i>	10483	10483	10483	752	752	752

